Euclid (*c.*330–*c.*260 BCE) Prominent Greek mathematician, often considered the father of geometry. A teacher at the mathematical school in Alexandria, Euclid is best known for his 13-volume treatise on geometry, *Elements*. It is considered the most important mathematical textbook of antiquity and was still in general use until the 19th century.

Euler, Leonhard, Swiss mathematician (1703–83) See p.152

Fabricius, Hieronymus, Italian surgeon (1537–1619) See p.93

Fahrenheit, Gabriel Daniel (1686–1736) German physicist and engineer who invented alcohol and mercury thermometers. Fahrenheit worked as a glassblower and chemistry lecturer in the Netherlands, where he also manufactured barometers, altimeters, and thermometers. In addition to developing the Fahrenheit scale, he discovered that water can remain a liquid below its freezing point.

Falloppio, Gabriele, Italian anatomist (1523–62) See p.83

Faraday, Michael, English chemist and physicist (1791–1867) See p.192

Fermat, Pierre de, French mathematician (1601–65) See p.104

Fermi, Enrico (1901–54) Italian physicist best known for developing atomic energy. A professor of theoretical physics at the University of Rome, Fermi was awarded the 1938 Nobel Prize in Physics for his work in induced radioactivity. A leading figure in the US's Manhattan Project to build an atomic bomb, he later designed the country's first nuclear reactor.

Feynman, Richard (1918–88) American physicist and co-winner of the 1965 Nobel Prize in Physics for developing quantum electrodynamics—the theory of the interaction between light and matter. He also created pictorial representations (Feynman diagrams) of interacting particles, provided an explanation of the physics of supercooled liquid helium, and contributed to the Manhattan Project.

Fibonacci, Leonardo, Italian mathematician (*c.*1170–*c.*1250) See p.59

Flamsteed, John (1646–1719) The first Astronomer Royal of England and who helped establish the Greenwich Observatory in London. Educated at Cambridge University and ordained a clergyman, Flamsteed is noted for his 1725 *Historia Coelestis Britannica*, which cataloged 3,000 stars. His observational data helped Isaac Newton verify his gravitational theory.

Fleming, Alexander (1881–1955) Scottish bacteriologist and co-winner of the 1945 Nobel Prize in Physiology or Medicine for discovering penicillin. He is also noted for discovering the antiseptic properties of the enzyme lysozyme, and for being the first to use antityphoid vaccines on humans.

Florey, Howard Walter (1898–1968) Australian pathologist who collaborated with Ernst Boris Chain to purify, isolate, and produce penicillin for medical use, for which both scientists were awarded the 1945 Nobel Prize in Physiology or Medicine. The manufacture of penicillin began in 1943, and saved the lives of countless war casualties.

Fossey, Dian (1932–85) American zoologist famous for her 18-year-long study of mountain gorillas in Rwanda. Anthropologist Louis Leakey convinced Fossey to undertake the study, which she began in 1967. She lived reclusively among gorillas and became a leading authority on their behavior. Fossey was murdered in 1985, after ensuring worldwide media coverage of the issue of gorilla poaching.

Foucault, Jean Bernard Leon (1819–68) French physicist famous for measuring the speed of light and showing that it travels more slowly through water than air. Foucault is also noted for inventing the gyroscope and using a giant pendulum to demonstrate that Earth rotates on its axis.

Fourier, Joseph, French mathematician (1768–1830) See p.183

Franklin, Benjamin, American inventor and scientist (1706–90) See p.143

Franklin, Rosalind, British chemist and biophysicist (1920–58) See p.283

Fraunhofer, Joseph von (1787–1826) German physicist who discovered the dark lines of the Sun's spectrum, now known as Fraunhofer lines, which later helped reveal the chemical composition of the Sun's atmosphere. To observe the lines, Fraunhofer designed and constructed achromatic lenses of high magnitude. He is considered the founder of the German optical industry.

Fresnel, Augustin Jean, French engineer (1788–1827) See p.179

Freud, Sigmund (1856–1939) Austrian neurologist and founder of psychoanalysis. Freud's methods advocated dialogue and "free association" to interpret childhood dreams, recollections, and infantile sexuality. Always controversial, Freud's ideas gained importance after World War I, especially in the US, but in 1933, Hitler banned psychoanalysis and Freud fled to England.

Gabor, Dennis (1900–79) Hungarian–British engineer and physicist, awarded the 1971 Nobel Prize in Physics for inventing holography, a method of 3-D photography. Originally a research engineer in Berlin, Gabor moved to London in 1933, where he worked on optics, oscilloscopes, and television. Holograms could not be produced until lasers were invented.

Galen, Claudius, Roman physician, surgeon, and philosopher (*c.*130–*c.*210) See p.37

Galilei, Galileo, Italian natural philosopher, astronomer, and mathematician (1564–1642) See p.97

Galvani, Luigi (1737–98) Italian physiologist who discovered that he could make the muscles of a dead frog twitch by applying two pieces of metal to nerve endings in its leg. This showed that nervous messages are carried by what was called "animal electricity," later shown to be the same as the electricity produced by a battery. Galvanization, or rust prevention, is named after Galvani.

Gamow, George (1904–68) Russian-born American nuclear physicist and cosmologist who helped develop the Big Bang theory of creation. He also correctly proposed that patterns within DNA form a genetic code. Gamow authored popular science books, including the notable *Mr. Tompkins* series.

Gassendi, Pierre (1592–1655) French priest, mathematician, and philosopher who tried to reconcile an atomic theory of matter based on Epicureanism with Christian doctrine. Gassendi took the harmony of nature as proof of the existence of God and is noted for his 1642 *Objections to Descartes' Meditations*. He was the first to observe the planetary transit of Mercury in 1631.

Gauss, Carl Friedrich, German mathematician and physicist (1777–1855) See p.163

Gay-Lussac, Joseph-Louis (1778–1850) French chemist and physicist noted for his investigations into gases. An assistant to chemist Berthollet, Gay-Lussac conducted experiments on gases, vapors, temperature, and terrestrial magnetism, sometimes from an ascending hot-air balloon. He discovered the law of combining volumes of gases as well as the element boron.

Geiger, Hans (1882–1945) German physicist who developed the Geiger counter for detecting and measuring radioactivity. Working under Ernest Rutherford at Manchester University, Geiger and Ernest Marsden undertook an experiment to show that an atom has a nucleus. He later worked with his student Walther Müller to improve the sensitivity of his Geiger counter.

Gilbert, William (1544–1603) English physicist and royal physician often regarded as the father of magnetic studies. Held in high esteem by his contemporaries, Gilbert was the first to establish the magnetic nature of Earth, and to use the terms: electric attraction, electric force, and magnetic pole.

Goddard, Robert H. (1882–1945) American physicist and inventor who created the first liquid-fueled rocket. A professor at Clark University, Goddard wrote *A Method of Reaching Extreme Altitudes* (1919)—considered a classic treatise on 20th-century rocket science. He developed three-axis control, gyroscopes, and steerable thrust for rockets, and successfully launched 34 rockets between 1926 and 1941.

Goeppert-Mayer, Marie (1906–72) German-born American theoretical physicist awarded the 1963 Nobel Prize in Physics for proposing the shell theory of nuclear structure. Goeppert-Mayer is also noted for her work in quantum electrodynamics and spectroscopy, and for researching organic molecules with her husband, American chemist Joseph Mayer. She also worked on the separation of uranium isotopes for the Manhattan Project.

Golgi, Camillo (1843–1926) Italian biologist, pathologist, and co-winner of the 1906 Nobel Prize in Physiology or Medicine for his investigations into the central nervous system. Golgi's development of a silver nitrate nerve-tissue staining technique called the "black reaction," allowed him to discover a connecting nerve cell, known as the Golgi cell.

Goodall, Jane (1934–) English ethologist best known for her 45-year study on the chimpanzees of Gombe Stream National Park, Tanzania. A one-time assistant to anthropologist Louis Leakey, Goodall established her Gombe Stream camp in 1960. She discovered that chimpanzees are omnivores, capable tool-makers, and have highly complex social behaviors.

Gould, Stephen Jay (1941–2002) American paleontologist and evolutionary biologist best known for creating the theory of punctuated equilibrium with Niles Eldredge. This theory proposes that evolution undergoes periods of relative stability, punctuated by short bursts of change. A Harvard University professor and popularizer of evolutionary theory, Gould campaigned against creationism and argued that science and religion be kept as two distinct fields.

Greene, Brian (1963–) American physicist, mathematician, and advocate of string theory, which tries to reconcile relativity and quantum theory, and proposes that minuscule strands of energy are responsible for creating every particle and force in the Universe. A well-known popularizer of science, his best-selling books include Pulitzer Prize finalist *The Elegant Universe*.

Guericke, Otto von (1602–86) German physicist, engineer, philosopher, and mayor of Magdeburg. Guericke invented the air pump, with which he was able to investigate atmospheric pressure and the properties of the vacuum, which he demonstrated to Emperor Ferdinand III. In 1663, Guericke produced static electricity by rubbing a spinning sulfur globe.

Gutenberg, Johannes, German inventor (*c.*1395–*c.*1468) See p.69

Guth, Alan (1947–) American theoretical physicist, cosmologist, and creator of the inflationary universe theory, which states that a rapid period of inflation during the Big Bang caused the Universe to expand exponentially—from microscopic to cosmic.

Haber, Fritz (1868–1934) German chemist awarded the 1918 Nobel Prize in Chemistry for synthesizing ammonia—an essential component of explosives and fertilizers. Together with Carl Bosch, Haber developed a process for the mass production of ammonia for use in fertilizer, a method still widely used today. Known as the father of chemical warfare, Haber also developed poisonous gases for use in World War I.

Hadley, George (1685–1768) English physicist and meteorologist whose theory of the trade winds explained why Northern Hemisphere winds blow from the north, and Southern Hemisphere winds blow from the southeast. Now known as Hadley's principle, it remained unacknowledged from 1735 until its rediscovery by John Dalton in 1793.

Haeckel, Ernst (1834–1919) German zoologist and Darwinist who was the first to map a genealogical tree relating all forms of life. A professor at Jena University, Germany, Haeckel studied marine organisms, described and named thousands of new animal species, and created the now discarded recapitulation theory, summarized as "ontogeny recapitulates phylogeny" (evolution can be seen in embryonic development).

Hahn, Otto (1879–1968) German chemist and pioneer of radioactivity and radiochemistry. Hahn's first great breakthrough came in 1917 when he and colleague Lise Meitner discovered the radioactive element protactinium. This was followed by the discovery of nuclear fission in 1938, for which he won the 1944 Nobel Prize in Chemistry. Hahn later became an outspoken opponent of nuclear weapons.